# AN AI ECOSYSTEM FOR FAKE JOB SCAM DETECTION AND FRAUD-AWARE JOB RECOMMENDATION

## COMMON INTEGRATED SOLUTION REPORT


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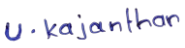
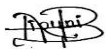
Sri Lanka


October 2026

# DECLARATION AND SUPERVISOR CERTIFICATION

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# ABSTRACT

Online recruitment fraud harms job seekers at several connected stages: the advertisement, the employer behind it, and the messages that follow. Existing work treats these stages separately, and job recommenders rank by relevance without considering fraud risk. This research presents an integrated AI ecosystem with four components. A multilingual fake job post detector (fine-tuned XLM-RoBERTa) classifies English and Sinhala advertisements with LIME explanations. An employer legitimacy verifier combines a Random Forest with registry, reputation and website evidence into a 0–100 score and a Low, Medium or High-risk band. A scam communication detector (DistilBERT, BSTD) classifies messages and identifies four manipulation tactics. A risk-aware ranking component combines TF-IDF skill relevance with the three risk values using the Entropy Weight Method and TOPSIS. The components run as separate Python services behind a Node.js backend, a recruiter web application and a job-seeker mobile application. On held-out data, the detectors reached 88.5% accuracy (post-detector, earlier data version), 87.0% accuracy and 78.1% fraud recall for the explainable employer score, and 85.3% binary accuracy with a tactic macro F1 of 0.835. Risk-aware ranking kept Precision@10 and MRR@10 unchanged for 50 test users while lowering average risk@10 from 0.1006 to 0.0900 and the unsafe rate from 2.80% to 2.40%. Four users tested the applications and found them understandable and useful. The evidence is technical and preliminary: the datasets are limited, the user study is small, and real-world fraud prevention was not measured. The value of the complete solution is that one job seeker query is answered with evidence from every stage of the recruitment process.



**Keywords:** recruitment fraud, fake job detection, employer verification, scam communication, risk-aware recommendation

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# TABLE OF CONTENTS

1 INTRODUCTION AND PROJECT CONTEXT ..... 1
1.1 Background and Context ..... 1
1.2 Stakeholders and user needs ..... 2
1.3 Alignment with the Sustainable Development Goals ..... 2
1.4 Purpose and scope of the common report ..... 3
2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS ..... 4
2.1 Overall problem definition ..... 4
2.2 Root causes and overall research gap ..... 4
2.3 Subproblem and component mapping ..... 5
2.4 Overall aim and group objectives ..... 6
2.5 Scope assumptions and constraints ..... 7
3 OVERALL SOLUTION STRUCTURE ..... 9
3.1 Integrated solution overview ..... 9
3.2 High-level architecture ..... 10
3.3 Component 1 summary ..... 11
3.4 Component 2 summary ..... 12
3.5 Component 3 summary ..... 13
3.6 Component 4 summary ..... 14
3.7 End to end user and data flow ..... 14
4 COMPONENT INTEGRATION AND COHESIVE OPERATION ..... 16
4.1 Integration strategy ..... 16
4.2 Interfaces data contracts and shared models ..... 17
4.3 Shared nonfunctional requirements ..... 18
4.4 Integrated workflow and exception handling ..... 18
4.5 Team contribution and dependency matrix ..... 19
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION ..... 21

5.1 Common technology stack and infrastructure ..... 21
5.2 Development and integration process ..... 21
5.3 Deployment and operational configuration ..... 22
5.4 Evidence of team execution ..... 23
6 INTEGRATED TESTING EVALUATION AND VALIDATION ..... 24
6.1 Evaluation framework and success criteria ..... 24
6.2 Integration and end to end testing ..... 25
6.3 Overall results ..... 25
6.4 Stakeholder and user validation ..... 28
6.5 Achievement of common objectives ..... 29
6.6 Integrated limitations and threats ..... 29
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION ..... 31
7.1 Value proposition and beneficiaries ..... 31
7.2 Market alternatives and differentiation ..... 31
7.3 Business model and feasibility ..... 32
7.4 Adoption intellectual property and compliance ..... 33
7.5 Sustainable Development Goal contribution ..... 33
8 CONCLUSION AND FUTURE DIRECTIONS ..... 35
8.1 Overall conclusion ..... 35
8.2 Principal integrated contribution ..... 35
8.3 Future directions ..... 36
REFERENCES ..... 38
APPENDICES ..... 40
Appendix A Integrated interface specification ..... 40
Appendix B Team contribution declaration ..... 41

# LIST OF FIGURES

Figure 2.1: The four stages of recruitment fraud and the component that addresses each ..... 5
Figure 3.3.1: High-level architecture of the integrated solution ..... 11
Figure 3.2: End-to-end flow of a job and a user through the system ..... 15
Figure 6.1: Average risk@10 and unsafe rate@10 for the baseline and the risk-aware ranking..... 27

# LIST OF TABLES

Table 1.1: Stakeholders and the needs addressed by the system ..... 2
Table 2.1: The four subproblems and why each is necessary ..... 5
Table 2.2: Group objectives and the evidence used to evaluate them ..... 6
Table 3.1: Summary of the four components ..... 9
Table 4.1: Integration risks and how they are handled ..... 16
Table 4.2: Interfaces between the components and the rest of the system ..... 17
Table 4.3: Shared non-functional requirements and the evidence for them ..... 18
Table 4.4: Failure paths and responses ..... 19
Table 4.5: Team contribution and dependency matrix ..... 20
Table 5.1: Common technology stack ..... 21
Table 5.2: Sources of evidence of team execution ..... 23
Table 6.1: Evaluation framework ..... 24
Table 6.2: Testing carried out and its results ..... 25
Table 6.3: Principal result of each component ..... 26
Table 6.4: Skill-only baseline compared with risk-aware ranking (50 users, K = 10) ..... 26
Table 6.5: Routing of the 300 test employers by risk band ..... 27
Table 6.6: Summary of user acceptance feedback ..... 28
Table 6.7: Achievement of the group objectives ..... 29
Table 6.8: Integrated limitations and threats ..... 29
Table 7.1: Existing alternatives and the proposed differentiation ..... 31
Table 7.2: Possible business models (not yet tested) ..... 32
Table 8.1: Future directions in priority order ..... 36
Table A.1: Component 2 interface ..... 40
Table B.1: Contribution declaration ..... 41

# LIST OF ABBREVIATIONS

| Abbreviation | Meaning |
|---|---|
| AI | Artificial Intelligence |
| AP@10 | Average Precision at 10 |
| API | Application Programming Interface |
| APK | Android Application Package |
| AUC | Area Under the Curve |
| BSTD | Behaviour-Inspired Semantic Tactic Detection |
| CI | Continuous Integration |
| CSE | Colombo Stock Exchange |
| eROC | Electronic Registrar of Companies portal |
| EWM | Entropy Weight Method |
| FOMO | Fear of Missing Out |
| HTTPS | Hypertext Transfer Protocol Secure |
| LIME | Local Interpretable Model-agnostic Explanations |
| MRR@10 | Mean Reciprocal Rank at 10 |
| NDCG@10 | Normalized Discounted Cumulative Gain at 10 |
| OCR | Optical Character Recognition |
| ROC-AUC | Area Under the Receiver Operating Characteristic Curve |
| SDG | Sustainable Development Goal |
| SLIIT | Sri Lanka Institute of Information Technology |
| SME | Small and Medium Enterprise |
| TF-IDF | Term Frequency–Inverse Document Frequency |
| TOPSIS | Technique for Order Preference by Similarity to Ideal Solution |
| UAT | User Acceptance Testing |

# 1 INTRODUCTION AND PROJECT CONTEXT

## 1.1 Background and Context

### 1.1.1 Recruitment fraud as a problem

Online recruitment has become a main way for job seekers to find work, and it has also become a channel for fraud. A recruitment scam usually starts with a deceptive job advertisement, may be backed by a company that does not exist or has no real record, and then continues in private messages where the scammer uses urgency and other pressure to obtain money or personal data [1], [2].

### 1.1.2 What existing research covers

Research on this problem has mostly classified the text of the advertisement as real or fake on public collections such as EMSCAD [1], [3], [4]. Such a classifier judges the advertisement and not the organization behind it, and it cannot see what happens after the job seeker replies.

### 1.1.3 Employer evidence and job recommendation

Employer-level evidence adds a second kind of signal: whether the company appears in official registers, on review platforms and on a working website. Studies that added such features to a classifier reported improvements in detection [5]–[7]. A third body of work recommends jobs to candidates by matching skills, and it ranks by relevance alone without asking whether the job is safe [8], [9].

### 1.1.4 The Sri Lankan context and the proposed system

For a Sri Lankan job seeker, the problem is sharper because many advertisements circulate in Sinhala and on social media, and the official registers that can confirm an employer are published in different places. This research therefore builds one system that checks the advertisement, the employer and the conversation, and then uses the three results when it recommends jobs.

## 1.2 Stakeholders and user needs

### 1.2.1 Stakeholders and what they need

The primary stakeholders are job seekers, who need relevant jobs, a way to tell whether an advertisement, an employer, or a message is risky, and an explanation they can act on. Legitimate recruiters need to be distinguished from fraudulent ones, and platform administrators need a way to decide quickly which registering companies need a person to look at them. Table 1.1 lists the stakeholders and what the system offers each of them.

Table 1.1: Stakeholders and the needs addressed by the system

| Stakeholder | Need | How the system responds |
|---|---|---|
| Job seekers (students, recent graduates, people searching online) | Relevant jobs that are also safe; to know why something is risky | Ranked jobs with relevance and safety; risk bands with reasons; warnings about manipulation tactics in messages; a public check of any company |
| Legitimate recruiters and employers | To be distinguished from fraudulent employers | Evidence-based employer score; automatic verification of Low and Medium risk registrations |
| Platform administrators | To review only the registrations that need attention | High-risk registrations are passed to an administrator together with the evidence report |
| Project supervisors and the external validator | Evidence that the research is sound and the product works | Component results, integrated tests and the acceptance sessions of Section 6.4 |

### 1.2.2 Basis of the needs

These needs were derived from the problem analysis and the system design, and they were reviewed in the user acceptance sessions with two recruiters and two job seekers (Section 6.4). No larger survey of stakeholders was carried out, so the needs should be read as reasoned and partly confirmed, not as the result of a formal needs study.

## 1.3 Alignment with the Sustainable Development Goals

### 1.3.1 Alignment with SDG 8

The project is aligned with United Nations Sustainable Development Goal 8, which promotes sustained, inclusive and sustainable economic growth, productive employment and decent work for all. Target 8.5 concerns full and productive

employment and decent work for all women and men, including young people [10]. By helping job seekers recognize fraudulent advertisements, employers and messages, and by ranking safer jobs higher, the system supports more informed job searches. The multilingual detector also supports Sinhala readers, who are often poorly served by English-only tools.

### 1.3.2 Nature of the contribution

The contribution is intended and indirect. The evaluation measures the performance of the components and the safety of ranked lists; it does not show that employment rose or that fraud losses fell. Section 7.5 discusses the possible negative effects and what would be needed to measure the contribution.

## 1.4 Purpose and scope of the common report

### 1.4.1 Purpose of the report

This common report presents the integrated problem and solution: how the four components fit together, how they were integrated and tested as one system, what the combined evidence shows, and what value the solution offers.

### 14.2 Relation to the individual theses

The detailed research and development of each component, including its dataset, model design, experiments and error analysis, are given in the four individual theses, and this report summarizes them only as far as needed to understand the whole. The report covers implemented functionality and measured results; proposed extensions and unverified benefits are identified as such.

### 1.4.3 Structure of the report

Chapter 2 defines the problem and the four subproblems. Chapter 3 presents the solution structure and the four components. Chapter 4 describes the integration, Chapter 5 the implementation and project execution, and Chapter 6 the testing, results and validation. Chapter 7 covers commercialization, sustainability and responsible innovation, and Chapter 8 gives the conclusion and future directions.

# 2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

## 2.1 Overall problem definition

### 2.1.1 A multi-stage problem

Online recruitment fraud is a multi-stage problem. A job seeker meets it first as an advertisement, then as an employer and then as a conversation, and each stage can be faked independently of the others. A convincing advertisement can come from a company that does not exist, a real company name can be used by someone else, and a harmless advertisement can lead to a conversation that demands payment. At the same time, jobs are recommended to candidates mainly by how well their skills match, and fraud assessments, where they exist, are not used in that choice [1], [8], [9].

### 2.1.2 The common problem

The common problem is therefore the lack of an integrated, risk-aware approach: how to assess recruitment fraud at the advertisement, employer and conversation stages, and how to use the resulting risk when recommending jobs, so that a job seeker is shown opportunities that are both relevant and safe.

## 2.2 Root causes and overall research gap

### 2.2.1 Limitations of current practice

Four limitations in current practice explain why one component is not enough. First, classifying the advertisement does not show whether the organisation behind it exists, so a well-written advertisement from a fake employer can pass [3], [4]. Second, employer checks that look at one registry or one platform are tied to the registers of one country and give no account of what each source contributed [5]–[7]. Third, labelling a message only as scam or not scam does not tell the user which manipulation tactic is being used, and fraud that happens in conversation is not covered by advertisement or employer checks [2]. Fourth, recommenders that rank by skill similarity do not take any of these risks into account [8], [9].

### 2.2.2 The overall research gap

The overall gap is the absence of a single framework that connects the four functions and passes their outputs to the ranking step in a form the ranking can use. Closing it needs more than four separate models: the outputs must describe the same job, share a scale, and be combined in a way that can be explained. Figure 2.1 shows how the stages of the problem map to the components.

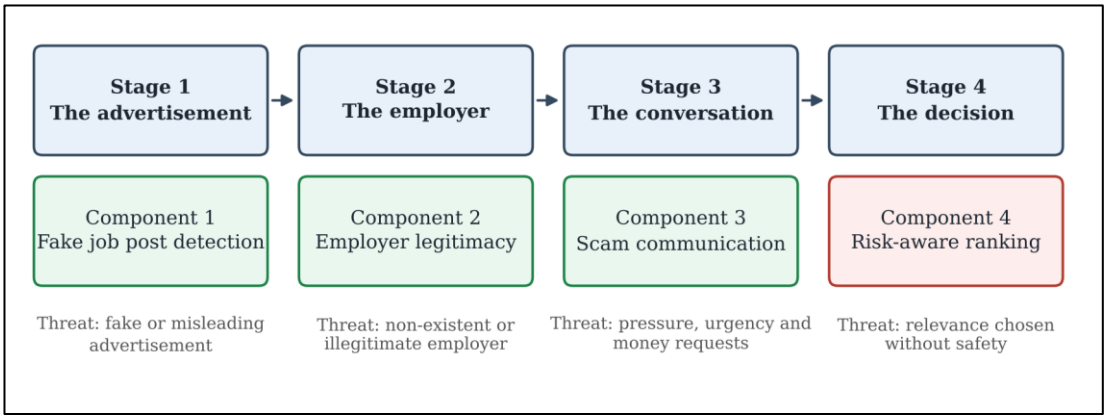



Figure 2.1: The four stages of recruitment fraud and the component that addresses each

## 2.3 Subproblem and component mapping

### 2.3.1 The four subproblems

The problem is divided into four subproblems that do not overlap but depend on one another (Table 2.1). Components 1 to 3 each produce a risk value for one stage. Component 4 uses all three and is therefore the point at which the system acts on the evidence.

Table 2.1: The four subproblems and why each is necessary

| Component and owner | Subproblem | Why it is needed on its own terms | What the others cannot replace |
|---|---|---|---|
| 1. Multilingual fake job post detection (Santhosh S) | Is this advertisement, in English or Sinhala, fraudulent? | The advertisement is what the job seeker sees first, and it can be checked before any contact | A genuine-looking employer can post a fake advertisement; an employer check does not read the text |
| 2. Employer legitimacy verification (Bandara H M N T) | Does the organization exist and does the evidence about it look genuine? | A company can be checked against registers, reviews and its website, independently of one advertisement | Advertisement text cannot show whether the company exists |

| Component and owner | Subproblem | Why it is needed on its own terms | What the others cannot replace |
|---|---|---|---|
| 3. Scam communication detection (Kajanthan U) | Is this message a scam, and which manipulation tactic does it use? | Fraud often moves to private messages after the advertisement | Neither the advertisement nor the employer check sees the conversation |
| 4. Risk-aware job recommendation and ranking (Methmi N P C S) | How should jobs be ranked when relevance and safety both matter? | Detection is useful to a job seeker only when it changes what is shown | Detectors alone produce warnings, not a decision about which jobs to show |

### 2.3.2 How the components answer the problem

Together the components answer the common problem. The first three examine different stages of the same scam, and the fourth turns their evidence into a ranked list. In the evaluation of Component 4, the risk-aware ranking kept Precision@10 and MRR@10 unchanged, slightly improved AP@10 and NDCG@10, and reduced average risk@10 from 0.1006 to 0.0900 and the unsafe rate@10 from 2.80% to 2.40% (Section 6.3).

## 2.4 Overall aim and group objectives

### 2.4.1 Aim and group objectives

The overall aim is to develop and evaluate an integrated AI-based ecosystem that detects recruitment fraud at the advertisement, employer and conversation levels, and that uses the resulting risk values in job recommendation to balance relevance and safety. This aim is derived from the group's research design and implemented evaluation. Table 2.2 gives the group objectives, the component responsible and the evidence used to judge each one.

Table 2.2: Group objectives and the evidence used to evaluate them

| Group objective | Component | Evidence and measure |
|---|---|---|
| G1. Detect fraudulent English and Sinhala job advertisements and explain the result | 1 | Accuracy, ROC-AUC and weighted F1 on a held-out test set; explanation by LIME |
| G2. Assess employer legitimacy from registry, reputation and website evidence | 2 | Accuracy, F1 and fraud recall on 300 held-out employers; fraud found in each risk band |
| G3. Detect scam messages and identify manipulation tactics | 3 | Binary accuracy and multi-label tactic macro F1 on held-out messages |

| Group objective | Component | Evidence and measure |
|---|---|---|
| G4. Rank jobs using both skill relevance and fraud risk | 4 | Precision@10, AP@10, MRR@10, NDCG@10 and risk and safety measures against a skill-only baseline for 50 users |
| G5. Operate the four components as one system | All | Integrated workflow tests, interface checks and user acceptance sessions (Sections 6.2 and 6.4) |

### 2.4.2 What the objectives do and do not show

The objectives are judged on technical evidence from datasets and test users. They do not establish the rate at which real scams are prevented, and the improvement in the ranking metrics is small and was not shown to be statistically significant.

## 2.5 Scope assumptions and constraints

### 2.5.1 Scope

The scope covers four connected functions: English and Sinhala fake job post detection, employer legitimacy assessment for Sri Lankan employers, scam communication classification with four manipulation tactics, and job ranking by relevance and risk. The system is delivered as a recruiter web application, a job-seeker mobile application, a Node.js backend and four Python services.

### 2.5.2 Assumptions and constraints

The following assumptions and constraints apply.

Language: advertisements in English and Sinhala are supported; Tamil is not.

Data: the post detector was built on 4,590 posts, of which 3,698 are legitimate and 892 fake (866 of the fake posts from EMSCAD); the employer evidence for 1,514 employers includes some feature values that were generated at fixed rates; the recommender was evaluated on 50 test users and 516 candidate records.

Sources: employer checks use public registers, review platforms and websites, which can change, limit automated access, or be unavailable.

Dependencies: the ranking component needs the risk values of the other three components for the same job.

Deployment: the web application is hosted on Vercel and the backend on Railway; the mobile application is built as an APK for demonstration and is not released through an application store.

Exclusions: production-scale load, live fraud-prevention outcomes, and a formal accessibility or security audit are outside the scope of this report.

# 3 OVERALL SOLUTION STRUCTURE

## 3.1 Integrated solution overview

### 3.1.1 Parts of the system

The solution is one system with two user-facing applications, a backend and four research services. A recruiter or administrator uses a web application to manage a profile, publish vacancies and review registrations. A job seeker uses a mobile application to browse jobs, receive ranked recommendations, check a company, and check messages. The backend stores users, jobs, applications, conversations and notifications, and it calls the four Python services that carry the research.

### 3.1.2 Central value

The central value is that one request from a job seeker is answered with evidence from every stage of the recruitment process, and that this evidence decides which jobs are shown first, instead of fraud checking and job matching being separate tools.

### 3.1.3 Comparison of the four components

Table 3.1 compares the four components on the same attributes, and Sections 3.3 to 3.6 give the details that matter for integration. The individual theses contain the full methods.

Table 3.1: Summary of the four components

| Attribute | Component 1 | Component 2 | Component 3 | Component 4 |
|---|---|---|---|---|
| Owner | Santhosh S | Bandara H M N T | Kajanthan U (team leader) | Methmi N P C S |
| Subproblem | Fake job post detection, English and Sinhala | Employer legitimacy | Scam communication and manipulation tactics | Risk-aware job ranking |
| Input | Advertisement text, or text read from a poster image | Company name, website, email | Message text, or text read from a screenshot | Candidate skills, job descriptions, three risk values |
| Core method | Fine-tuned XLM-RoBERTa; LIME | Random Forest on 14 features; weighted fusion of registry, reputation and website evidence | Fine-tuned DistilBERT (BSTD) | TF-IDF and cosine similarity; EWM; TOPSIS |
| Output | Legitimate, suspicious or fake; explanation | Score 0–100, band Low, Medium or High, evidence report | Scam or not; four tactics; warnings | Ranked jobs with relevance, safety, matched skills, skills to develop |

| Attribute | Component 1 | Component 2 | Component 3 | Component 4 |
|---|---|---|---|---|
| Role in the whole | Advertisement risk | Employer risk, r = 1 − S/100 | Conversation risk | Combines the three risks with relevance |
| Main result | 88.5% accuracy, ROC-AUC 0.924 (earlier data version) | 87.0% accuracy, 78.1% fraud recall (fused score); AUC 0.973 (model) | 85.3% binary accuracy; tactic macro F1 0.835 | Average risk@10 0.1006 to 0.0900 |

## 3.2 High-level architecture

### 3.2.1 Clients, backend and data stores

Figure 3.1 shows the architecture. The clients talk to the backend over HTTP and JSON. The backend, built with Node.js and Express and hosted on Render, uses MongoDB for application data, RabbitMQ for asynchronous events and Socket.IO for real-time chat.

### 3.2.2 Research services and external sources

The four components are separate FastAPI services, so each can be developed, tested and replaced on its own. Components 1 to 3 each return a risk value for one stage of the scam, and these three values go to Component 4, which returns a ranked job list. Component 2 also reads external public sources (the registers and review platforms of Section 3.4); these are outside the system boundary and are the main external dependency.

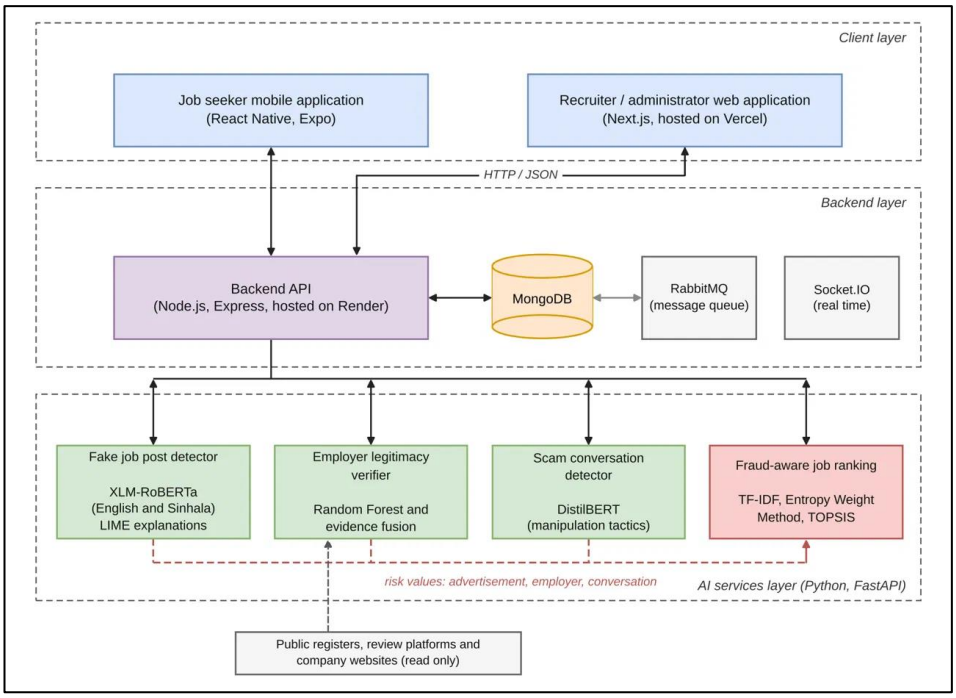



Figure 3.3.1: High-level architecture of the integrated solution

## 3.3 Component 1 summary

### 3.3.1 Purpose, input and output

Component 1, owned by Santhosh S, classifies a job advertisement as legitimate, suspicious, or fake. The input is the advertisement text, and text extracted by optical character recognition from poster-style images can also be analyzed. The output is a three-tier label with an explanation.

### 3.3.2 Method and data

The model is a fine-tuned XLM-RoBERTa [11], a multilingual transformer, so one model handles English and Sinhala. The component was built on 4,590 posts (3,698 legitimate and 892 fake, 866 of the fake posts from EMSCAD) with an 80/20 split. Explanations use LIME [12] as the main method, supported by leave-one-out token attribution, and they show which words pushed the decision.

### 3.3.3 Results and limits

In the first step, the held-out accuracy was 88.5%, the ROC-AUC 0.924 and the weighted F1 0.89 (635 test posts, earlier data version). The fake posts come mostly

from one public collection, so the result describes the data used and not every advertisement style.

### 3.3.4 Role in the ecosystem

The component supplies the advertisement risk indicator to the recommendation component, and its explanation lets the job seeker see why an advertisement was flagged (Section 3.7).

## 3.4 Component 2 summary

### 3.4.1 Purpose, input and output

Component 2, owned by Bandara H M N T, assesses the organization behind a vacancy. The input is the company name, with the website address and email address as optional fields. The output is a score S from 0 to 100, a band (Low risk at 70 or more, Medium at 45 to 69 and High below 45) and an evidence report that links each statement to its source.

### 3.4.2 Method

The component has three evidence layers. The registry layer searches the Colombo Stock Exchange, the Department of Registrar of Companies (eROC), the Central Bank, the Board of Investment, the Insurance Regulatory Commission and the Specified Business Enterprises list. The reputation layer looks for the company on review and professional platforms and for scam reports. The website layer checks HTTPS, About and Contact pages, privacy terms and signs of weak security. A Random Forest with 200 trees and depth 8 [13] turns 14 structured features into a model score, and this score is fused with the three-layer scores using the weights 0.40, 0.30, 0.20 and 0.10 for model, registry, reputation and website. A rule for small genuine companies lifts a clean company with no registry entry out of the High band.

### 3.4.3 Results and limits

On 300 held-out employers, the Random Forest reached an accuracy of 0.917 and an AUC of 0.973. The explainable fused score, which is the one shown to users, reached an accuracy of 87.0%, an F1 of 90.2% and a fraud recall of 78.1%. With the bands 45 and 70, none of the 107 employers in the Low band was fraudulent, 82 of the 105

fraudulent employers were in the High band and 23 in the Medium band. The training data include some feature values generated at fixed rates, so these figures show how the method behaves under controlled conditions and not its accuracy on all live employers.

#### 3.4.4 Role in the ecosystem

The component hands the other components one value, the employer risk $r = 1 - S/100$, and it screens company registrations on the platform: Low and Medium companies are verified automatically and High companies go to an administrator with the evidence report (Section 3.7).

### 3.5 Component 3 summary

#### 3.5.1 Purpose, input and output

Component 3, owned by Kajanthan U, analyses messages between a recruiter and a candidate. The input is message text, or text extracted from a screenshot of a conversation. The output is a scam label, the tactics found and explanatory warnings.

#### 3.5.2 Method

The Behavior-Inspired Semantic Tactic Detection (BSTD) framework uses a fine-tuned DistilBERT to classify a message as scam or not and to detect four manipulation tactics: urgency pressure, fear of missing out (FOMO), social proof and sunk-cost influence. The design follows work on BERT-based detection of fraudulent messages [14] and on reading text from message images [15], and adds the tactic layer, which tells the user not only that a message is suspicious but how it is trying to influence them.

#### 3.5.3 Results and limits

The binary classifier reached an accuracy of about 85.3% and the tactic detector a macro F1 of 0.8345, with FOMO the weakest of the four tactics.

#### 3.5.4 Role in the ecosystem

The component covers the last stage of the scam, the conversation, and supplies the conversation risk indicator to the recommendation component (Section 3.7).

## 3.6 Component 4 summary

### 3.6.1 Purpose, input and output

Component 4, owned by Methmi N P C S, ranks jobs for a candidate. The output lists each job with its relevance, safety and overall score, the matched skills and the skills to develop.

### 3.6.2 Method

Skill relevance is the cosine similarity between TF-IDF vectors of the candidate skills and the job description. The Entropy Weight Method gives a weight to each of the three risk indicators from the way the values vary across the jobs, so an indicator that separates jobs well counts for more; the weighted indicators give a risk score, and the safety score is its complement. TOPSIS then ranks the jobs on relevance and safety together, as distances to the ideal and the worst alternatives.

### 3.6.3 Results and limits

In the evaluation with 50 test users, 516 candidate records and a cutoff of 10, the risk-aware ranking was compared with a skill-only baseline (Section 6.3). Average risk@10 fell from 0.1006 to 0.0900 and the unsafe rate from 2.80% to 2.40%, while Precision@10 and MRR@10 did not change; the differences were small and not statistically significant.

### 3.6.4 Role in the ecosystem

The component is the point where the other three results meet: it combines the risk indicators of the advertisement, the employer and the conversation with skill relevance, and the job seeker sees the ranked list in the mobile application.

## 3.7 End-to-end user and data flow

### 3.7.1 Flow of a job and a user

Figure 3.2 shows how a job and a user move through the system. A recruiter publishes a vacancy in the web application and the backend stores it. The employer behind the vacancy is checked by Component 2 and the advertisement text by Component 1, and

the two risk values are stored with the job. When a job seeker opens the mobile application, the backend sends the candidate skills, the jobs and their risk values to Component 4, which returns a ranked list, and the application shows each job with its relevance, its safety and the reasons behind its risk. At any time, the job seeker can submit a message or a screenshot to Component 3, which returns a scam label and the tactics found, and can check any company by name and website through Component 2.

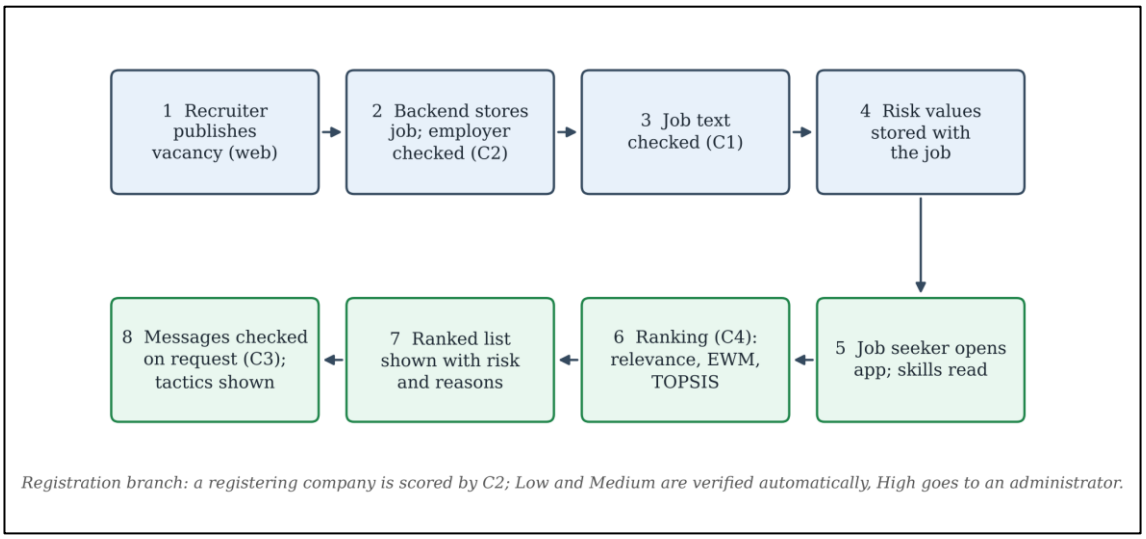



Figure 3.2: End-to-end flow of a job and a user through the system

### 3.7.2 Checking a company that registers

Component 2 has a second entry point. When a company registers as an employer on the platform, the registration is checked automatically after it is submitted. A company in the Low or Medium band is verified automatically, and a company in the High band is passed to an administrator together with the evidence report, so that a person can decide. Section 6.3 reports what this rule does on the test data, and Section 6.6 discusses the risk that comes with verifying the Medium band automatically.

# 4 COMPONENT INTEGRATION AND COHESIVE OPERATION

## 4.1 Integration strategy

### 4.1.1 Service-oriented integration

The integration is service-oriented. Each research component is a separate Python service with its own interface, and the Node.js backend and the applications call these services over HTTP. This keeps the components independent: each member developed and tested one service, and the team leader coordinated how they connect to the backend and the applications.

### 4.1.2 Order of integration and automated checks

The integration followed the order of the data: first vacancies and users, then the three detectors, and last the ranking component, which depends on all of them. Each service validates its input, reports errors in its response, and has automated tests, and a GitHub Actions workflow runs selected test suites when code changes (Section 5.2).

### 4.1.3 Integration risks

Table 4.1 lists the integration risks that this report identifies and how each one is handled. Some are handled in the code, and some are recommended actions that remain open. The report does not claim a formal risk register that was kept during the project.

Table 4.1: Integration risks and how they are handled

| Risk | Why it matters | How it is handled | Status |
|---|---|---|---|
| Different risk scales and meanings | Component 4 cannot combine values that mean different things | Each detector returns one risk value in [0, 1] for a job; Component 2 converts its score with r = 1 − S/100 | In place; scaling of the other services is described in the individual theses |
| A risk value does not match the job | A wrong match gives a wrong ranking | Risk values are stored with the job record and read from it by Component 4 | In place |
| A service is down, or a model is missing | The workflow could fail | Services report an error when a model is unavailable; Component 2 lowers its confidence when a website cannot be read | Partly; no system-wide fallback (Section 4.4) |
| Library version mismatch | A saved model may warn or fail when loaded with a different version | Versions recorded for Component 2 (scikit-learn 1.9.1, model saved with 1.9.0) | Open: pin versions in all requirements files |

| Risk | Why it matters | How it is handled | Status |
|---|---|---|---|
| Registers change or block access | Component 2 reads public pages | Layers fail independently; result gets a lower confidence | Partly; one unhandled registry error found |
| Automatic verification of a Medium-band company | About 22% of fraudulent test employers fall in Medium | Evidence reports and “not confirmed” wording; later review recommended | Open: Section 6.6 |

## 4.2 Interfaces data contracts and shared models

### 4.2.1 Interfaces and shared model

The services exchange JSON over HTTP, and the shared model is the job record and its risk values. Table 4.2 describes what each interface carries. Each service defines its own request and response models and validates them. The report does not claim one universal schema, a shared identifier standard, or a versioning policy for all services; the job identifier of the backend is what ties the risk values to a job.

Table 4.2: Interfaces between the components and the rest of the system

| Producer | Consumer | Content | Notes |
|---|---|---|---|
| Backend | Component 1 | Advertisement text, or an image for text extraction | Returns three-tier label, probabilities, explanation |
| Backend, mobile app | Component 2 | company_name, website_url, email (all optional) | POST /predict; returns risk_score S, risk_level, verdict, confidence, evidence report, elapsed_seconds |
| Backend, mobile app | Component 3 | Message text, or a screenshot for text extraction | Returns scam label, tactics, warnings |
| Backend | Component 4 | Candidate skills, jobs, three risk values per job | Returns ranked jobs with relevance, safety, overall score, matched skills, skills to develop |
| Components 1–3 | Component 4 (through the backend) | One risk value in [0, 1] per job and stage | Component 2: $r = 1 - S/100$ |

### 4.2.2 Error handling

Error handling is the responsibility of each service. A service returns an error response for invalid input or an unavailable model, and the caller decides what to show. Appendix A gives the interface of Component 2 in detail.

## 4.3 Shared nonfunctional requirements

Table 4.3 sets out the shared non-functional requirements, how the solution meets them and what evidence exists. The evidence is uneven, and the table says so.

Table 4.3: Shared non-functional requirements and the evidence for them

| Requirement | How it is met | Evidence | Status |
|---|---|---|---|
| Explainability | Word-level explanations for advertisements; an evidence report and score breakdown for employers; the tactics found in messages; matched skills and a safety score for jobs | Output fields of each service; acceptance sessions asked for clearer explanations | Met; wording to be improved |
| Reliability | Input validation, error responses and health checks in each service; the employer checks fail independently of one another | Service tests and automated checks; scenario tests of the employer service passed in most cases and exposed a few defects that remain open | Partly met |
| Performance | Parallel source checks and caching in the employer service; models loaded once at start-up | Model and scoring take only a few milliseconds per request with the external sources replaced; time with live sources not measured | Not established for the live system |
| Security | Authentication and authorization in the backend; settings read from environment variables | Tests of selected security functions; one default access key remains in the source code of a service | Partly met; no security audit |
| Privacy | The employer service stores no user data and no request history and uses organization-level data only | Design and code review | Met for the services; to be checked for the whole platform [16] |
| Usability | Web and mobile applications; short explanations | Four participants (Section 6.4) | Preliminary |
| Maintainability | Separate services with their own interfaces; version control; automated checks on every push | Repository structure; automated tests for every component | Met |
| Scalability and accessibility | Stateless services | No load test or accessibility assessment | Not evaluated |

## 4.4 Integrated workflow and exception handling

### 4.4.1 Normal workflow

In normal operation, the workflow is the one of Figure 3.2: a vacancy is published, its employer and text are checked, the three risk values are stored, and a ranked list is

returned to the job seeker. The team tested individual components and the full path from job publication to recommended jobs by hand.

### 4.4.2 Failure paths

Table 4.4 lists the important failure paths and what happens in each. Failures of one detector are handled inside that service. The system-wide behavior when a whole service is unavailable has not been defined or tested, and a fallback is recommended in Section 8.3

Table 4.4: Failure paths and responses

| Failure | Current behavior | Recommended |
|---|---|---|
| Invalid or missing input | The service validates the request and returns an error | Show a clear message in the application |
| Model file unavailable | The service reports an error | Health check before use; alert |
| Employer website cannot be read | Result returned with a low-confidence flag and a warning | Keep |
| A registry lookup raises an unhandled error | The request fails with a server error | Catch the error around each source and continue with lower confidence |
| Empty request to the employer service | Prediction Unknown, but the band is Medium with score 48 | Return no band and send the case to an administrator |
| A detector is unavailable when ranking | Not defined | Rank on relevance and mark the missing risk as unknown |
| Medium-band company verified automatically | Verified | Provisional verification, “not confirmed” note and later review |

## 4.5 Team contribution and dependency matrix

Table 4.5 maps each member to a component, the research and development contribution, and the dependency on the others. It shows the allocation of the four research components and the integration role of the team leader. The allocation of shared frontend, backend, deployment and documentation tasks to particular members is not recorded individually and is not assigned here; Appendix B records the contributions that all four members confirm.

Table 4.5: Team contribution and dependency matrix

| Member | Component | Main contribution | Depends on / supplies |
|---|---|---|---|
| Santhosh S (IT22113740) | 1. Multilingual fake job post detection | Dataset, XLM-RoBERTa model, explanation, evaluation | Supplies advertisement risk to the ranking component |
| Bandara H M N T (IT22251664) | 2. Employer legitimacy verification | Registry, reputation and website layers, Random Forest, fusion, evidence report, tests | Supplies employer risk to the ranking component; also public check and registration screening |
| Kajanthan U (IT22224002) | 3. Scam communication detection; team leader | BSTD model, tactic detection, evaluation; coordination of integration | Supplies conversation risk to the ranking component; coordinates the interfaces |
| Methmi N P C S (IT22188618) | 4. Risk-aware job recommendation and ranking | TF-IDF relevance, EWM risk aggregation, TOPSIS ranking, evaluation | Needs the risk values of the other three components; supplies the ranked list to the application |

# 5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

## 5.1 Common technology stack and infrastructure

The system combines web and mobile technologies, backend services, a database and machine-learning libraries (Table 5.1). The technologies were chosen to give each part the tool it needs: React Native with Expo for one codebase on the phone, Next.js for the recruiter web application, Node.js and Express for the business functions, and Python for the models, because the machine-learning libraries are available there. Keeping the models in separate FastAPI services means that a model or its libraries can change without touching the applications.

Table 5.1: Common technology stack

| Layer | Technology | Purpose |
|---|---|---|
| Mobile application | React Native, Expo | Job-seeker interface; built as an APK for demonstration |
| Web application | Next.js, React | Recruiter and administrator interface |
| Backend | Node.js, Express, Mongoose | Users, jobs, applications, chat, notifications; calls the research services |
| Data and messaging | MongoDB, RabbitMQ, Socket.IO | Application data, asynchronous events, real-time communication |
| Research services | Python, FastAPI | One service per component, with typed request models and generated interface documentation |
| Models and data | PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy | Transformer models (post and conversation detectors), Random Forest (employer verifier), data processing |
| Ranking | TF-IDF, cosine similarity, EWM, TOPSIS | Risk-aware ranking |
| Source control and CI | GitHub, GitHub Actions | Shared repository and automatically selected tests |

## 5.2 Development and integration process

### 5.2.1 Organization of development

Development was organized around the four research components and the applications that join them. Each member built and tested one component and its service interface, and the team leader coordinated the connection of the services to the backend and the applications.

### 5.2.2 Version control and automated tests

The source code is in a GitHub repository. A GitHub Actions workflow runs selected automated tests when changes are pushed or a pull request is opened against the main branches, and it separates the tests of the machine-learning components from those of the backend, so that the relevant tests run for the part that changed. The team tested the components and the integrated workflow from job publication to recommended jobs.

### 5.2.3 Practices not followed

The team did not keep a formal sprint schedule, a record of code reviews or a defined release procedure, and this report does not claim them. These practices are listed as improvements in Section 8.3.

## 5.3 Deployment and operational configuration

### 5.3.1 Web application, backend and services

The recruiter web application is hosted on Vercel and the backend on Railway, and the web application reaches the backend through configured service URLs. Settings and service addresses are read from environment variables. The research components run as Python services; for example, Component 2 is started with Uvicorn on its own port, loads its model once at start-up and adds the time taken to every response. MongoDB stores the application data.

### 5.3.2 Mobile application and operation

The mobile application is built as an Android Application Package (APK) and installed on a phone for demonstration, and it uses the addresses of the deployed backend and not development-only local addresses. This is a testing and demonstration arrangement and not a release through an application store. Health checks, input validation, error responses and automated tests support operation, but continuous production monitoring, a backup and restoration procedure and a disaster-recovery plan are not in place and are needed before a production release.

### 5.4 Evidence of team execution

The evidence that the work was carried out by the team is held in the places listed in Table 5.2. The raw logs are not reproduced here.

Table 5.2: Sources of evidence of team execution

| Source | What it shows |
|---|---|
| GitHub repository and history | The code of the applications, the backend and the research services, and the changes made to them |
| GitHub Actions runs | Selected automated tests run on pushes and pull requests |
| Automated test files | Unit and scenario tests kept in the repository for the research services |
| Microsoft Teams meetings | Team discussions and decisions, and the user acceptance sessions with two recruiters and two job seekers (Section 6.4) |
| Microsoft Planner | The plan of tasks, who was responsible for them and their timing |
| Log book | A record of the work done and the meetings held |
| The four individual theses | Component-level research, datasets, experiments and results |

# 6 INTEGRATED TESTING EVALUATION AND VALIDATION

## 6.1 Evaluation framework and success criteria

### 6.1.1 Evaluation questions and criteria

The evaluation asks two questions. First, does each component perform well enough on its own task? Second, does the complete system help a job seeker find relevant jobs while taking fraud risk into account? Table 6.1 lists the evidence, the measures and the criterion for each component. The criterion for the detectors is performance on data that was held out from training, judged against the weaknesses reported in the individual theses; the criterion for the ranking is that relevance is maintained while safety improves.

Table 6.1: Evaluation framework

| Component | Evidence | Measures | Success criterion |
|---|---|---|---|
| 1. Post detection | Held-out test posts (635 in the group paper) | Accuracy, ROC-AUC, precision, recall, F1 | Good separation of fake from legitimate posts on unseen posts |
| 2. Employer verification | 300 held-out employers (195 legitimate, 105 fraudulent), split by website domain | Accuracy, F1, fraud recall, ROC-AUC; fraud in each risk band | No fraudulent employer in the Low band; high fraud recall |
| 3. Scam communication | Held-out messages | Binary accuracy; macro F1 for four tactics | Reliable scam label and tactic identification |
| 4. Ranking | 50 test users, 516 candidate records, cutoff 10 | Precision@10, AP@10, MRR@10, NDCG@10; average risk@10, average safety@10, unsafe rate@10 | Relevance maintained; safety improved |
| Whole system | Integrated tests; four participants | Workflow completion; user feedback | Workflow runs end to end; users find it understandable and useful |

### 6.1.2 Comparability of the results

Results from different components come from different datasets and procedures, and they are not combined into one overall accuracy.

## 6.2 Integration and end to end testing

### 6.2.1 Levels of testing

Testing took place at five levels (Table 6.2). Unit tests check the functions of each research service, such as the scoring and model code and the handling of input. Scenario tests call a service through its HTTP interface with sources replaced by fixed inputs; they passed in most cases and exposed a small number of defects, which are listed in Section 6.6. The services are connected to the backend and the applications, and the path from publishing a vacancy to the recommended list in the mobile application was tested manually. A GitHub Actions workflow runs selected test suites automatically, and the applications were then evaluated by users in acceptance sessions. The detailed test cases and counts for each service are given in the individual theses.

Table 6.2: Testing carried out and its results

| Level | What was tested | Result |
|---|---|---|
| Unit tests | Scoring, model and input-handling functions of each research service | Carried out for every service; results in the individual theses |
| Scenario tests | Each service through its HTTP interface, including empty input, unreadable sources and an unavailable model | Passed in most cases; a small number of defects found and still open (Section 6.6) |
| Continuous integration | Selected test suites for the machine-learning services and the backend | Run on pushes and pull requests |
| End-to-end (manual) | Publish a vacancy, see it in the mobile application, receive recommendations | Workflow completed |
| User acceptance | Recruiter dashboard and job-seeker mobile application | Understandable and useful; clearer explanations wanted (Section 6.4) |

## 6.3 Overall results

### 6.3.1 Result of each component

Table 6.3 gives the main result and the main weakness of each component. Each component performs well on its own evaluation data, and each has a weakness that is stated openly.

Table 6.3: Principal result of each component

| Component | Evidence | Principal result | Main weakness |
|---|---|---|---|
| 1. Post detection | Held-out posts from a dataset of 4,590 posts | Accuracy 88.5%, ROC-AUC 0.924, weighted F1 0.89 (635 held-out posts) | Fake posts come mostly from one public collection; Sinhala coverage to improve |
| 2. Employer verification | 300 held-out employers | Accuracy 0.917 and AUC 0.973 for the model; accuracy 87.0% and fraud recall 78.1% for the explainable score; no fraud in the Low band | Some fraud reaches the Medium band |
| 3. Scam communication | Binary and multi-label evaluation | Binary accuracy about 85.3%; tactic macro F1 0.8345 | Fear of missing out is the weakest tactic |
| 4. Ranking | 50 users, Top-10 | Relevance maintained; average risk@10 down 10.5%; unsafe rate@10 down 14.3% | Gain is small and not statistically significant |

### 6.3.2 Ranking results

The ranking results are in Table 6.4 and Figure 6.1. Precision@10 and MRR@10 were unchanged, and AP@10 and NDCG@10 improved slightly. Average risk@10 fell from 0.1006 to 0.0900, a reduction of about 10.5%, and the unsafe rate@10 fell from 2.80% to 2.40%, about 14.3%. Average safety@10 rose from 0.8994 to 0.9100. The changes in the relevance measures are small and were not shown to be statistically significant, so the finding is that safety improved without a measured loss of relevance, and not that the ranking is proven to be better in live use.

Table 6.4: Skill-only baseline compared with risk-aware ranking (50 users, K = 10)

| Measure | Baseline | Risk-aware |
|---|---|---|
| Precision@10 | Reference | Equal to baseline |
| MRR@10 | Reference | Equal to baseline |
| AP@10 | Reference | Slightly above baseline |
| NDCG@10 | Reference | Slightly above baseline |
| Average risk@10 | 0.1006 | 0.0900 |
| Average safety@10 | 0.8994 | 0.9100 |
| Unsafe rate@10 | 2.80% | 2.40% |

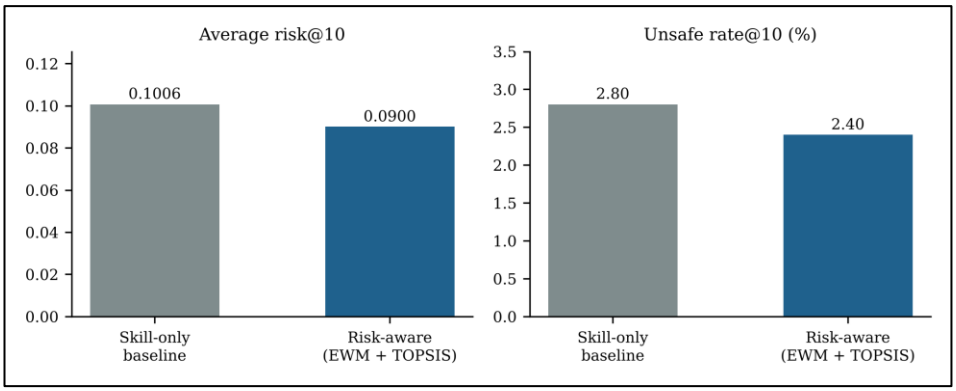



Figure 6.1: Average risk@10 and unsafe rate@10 for the baseline and the risk-aware ranking

### 6.3.3 Screening of company registrations

One system-level feature is the automatic screening of company registrations (Section 3.7), which sends companies in the High band to an administrator. Applied to 300 held-out employers (Table 6.5), the administrator would receive about one third of the registrations and most of the fraudulent ones, while no fraudulent company would reach the Low band. The Medium band is where an automatic decision can be wrong, and Section 6.6 returns to this.

Table 6.5: Routing of the 300 test employers by risk band

| Band | Genuine | Fraudulent | Total | Action |
|---|---|---|---|---|
| High (below 45) | 16 | 82 | 98 | Administrator reviews, with evidence report |
| Medium (45 to 69) | 72 | 23 | 95 | Verified automatically |
| Low (70 or more) | 107 | 0 | 107 | Verified automatically |
| Total | 195 | 105 | 300 | |

### 6.3.4 Overall assessment

Taken together, the results show that each part of the system works on its own held-out data, that combining the three risk signals in the ranking lowers the risk of the recommended jobs without a measured loss of relevance, and that automatic screening of registrations can reduce manual work. The evidence is limited by small and partly generated datasets, a small ranking gain, a four-person user study and the absence of live-system tests, so the results show that the approach works under controlled conditions and not yet at scale.

## 6.4 Stakeholder and user validation

### 6.4.1 Participants and feedback

User acceptance evaluation involved two recruiters, who tested the hosted recruiter dashboard, and two job seekers, who tested the mobile application installed as an APK. Feedback was collected through surveys and sessions held on Microsoft Teams. The sessions looked at usability, usefulness, and the value of the fraud-awareness functions. Table 6.6 summarizes the feedback.

Table 6.6: Summary of user acceptance feedback

| Group | Positive feedback | Suggestions |
|---|---|---|
| Recruiters (2) | Dashboard well organized; easy navigation; job management and candidate information easy to reach; fraud awareness valued because fake advertisements damage trust | More guidance for first-time users |
| Job seekers (2) | Application understandable; recommendations and safety information help decisions; checking advertisements, employers and messages is useful | Clearer explanations of risk assessments |

### 6.4.2 Interpretation of the feedback

Both groups found the applications understandable and useful. Recruiters valued the organized dashboard and the awareness of fake advertisements, and job seekers valued the safety information that helps them decide which jobs to pursue. This supports the applications' usability and the perceived usefulness of the fraud-aware functions, and it aligns with the system's aim of placing job risk alongside relevance.

The two suggestions point to concrete improvements. First, recruiters asked for more guidance for first-time users, which calls for short onboarding tips and a quick guide in the dashboard. Second, job seekers asked for clearer explanations of the risk assessments, which calls for plain-language reasons for each risk band and a statement of what the user should do next. The second suggestion also aligns with the explainability requirement in Table 4.3, whose wording is marked for improvement. The team will make both changes before the final viva, and will then repeat a short check with the participants to see whether the guidance and the explanations are understood; the outcome will be recorded in the log book.

The evidence has clear limits. Four participants are too few for a representative assessment, no task times or rating scores were recorded, and the sessions do not show wider market acceptance or better job-search outcomes. The finding is therefore a

preliminary confirmation of usability and usefulness and a source of specific improvements, and a larger and more varied study is listed as future work.

## 6.5 Achievement of common objectives

Table 6.7 relates the evidence to each group objective of Table 2.2.

Table 6.7: Achievement of the group objectives

| Objective | Evidence | Assessment |
|---|---|---|
| G1. Post detection for English and Sinhala | Accuracy 88.5%, ROC-AUC 0.924 on 635 held-out posts | Achieved within the evaluated data |
| G2. Employer legitimacy | Accuracy 87.0%, fraud recall 78.1%; Low band without fraud on the test split | Achieved within the evaluated data; Medium band is a weakness |
| G3. Scam messages and tactics | Binary accuracy 85.3%; macro F1 0.8345 | Achieved within the evaluated data; FOMO weakest |
| G4. Ranking by relevance and safety | Risk@10 and unsafe rate down; relevance unchanged or slightly higher | Partly achieved: safety improved, relevance changes small and not significant |
| G5. One operating system | Manual end-to-end test; acceptance sessions; CI | Achieved for the demonstrated workflow; failure handling and load not tested |

## 6.6 Integrated limitations and threats

Table 6.8 lists the limitations of the integrated solution, what each one affects and what should be done.

Table 6.8: Integrated limitations and threats

| Limitation | Effect on the findings | Action |
|---|---|---|
| Datasets are limited and partly generated (employer feature values; posts from EMSCAD and local sources) | Results show the methods under controlled conditions, not accuracy on all live recruitment traffic | Collect larger real data; repeat the evaluation |
| Medium band verified automatically | 23 of 105 fraudulent test employers (22%) would be verified without a person; 72 genuine employers sit in the same band | Verify provisionally, show “not confirmed”, keep the evidence report, review later |
| Open defects in the employer service | An empty request could be verified automatically; a registry error could stop a request | Correct before release |
| Small user study (4 participants) | Usability and acceptance are only preliminary | Larger study after ethical clearance |

| Limitation | Effect on the findings | Action |
|---|---|---|
| Ranking improvement is small and not significant | Cannot claim a proven advantage in live use | More users and a significance test |
| No load, accessibility, or security audit; no system-wide fallback | Reliability at scale is unknown | Systematic non-functional tests |
| Dependence on public registers and review platforms | Pages may change or block access | Seek published interfaces; monitor |
| FOMO detection is weaker (conversation detector) | Some subtle pressure is missed | More training data for FOMO |

The components depend on one another through the ranking component, so an error in one risk value affects the ranked list. A job seeker should see the risk values and their reasons next to each job and not only the position in the list.

# 7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

## 7.1 Value proposition and beneficiaries

### 7.1.1 Value and beneficiaries

The ecosystem offers safer online recruitment by joining four checks that are usually separate. Job seekers use the mobile application to explore jobs, to see a risk band and the reasons for it, to check a company before applying and to check messages, and recruiters use the web application to manage vacancies. The ranking puts safer relevant jobs first. The main beneficiaries are job seekers, especially students and recent graduates who search for work online. Legitimate employers' benefit when fraudulent ones are separated from them, and job platforms, educational institutions and public bodies interested in employment safety are other possible beneficiaries.

### 7.1.2 Evidence of demand and next steps

Feedback from two recruiters and two job seekers was positive about understandability and usefulness, but it does not establish market demand. The team intends to launch the product in the near future. Before that, demand has to be assessed, the user experience improved and the performance validated under real conditions; these are next steps and not completed commercial work.

## 7.2 Market alternatives and differentiation

### 7.2.1 Existing alternatives

The project compared the ecosystem with existing practices (Table 7.1). Each alternative covers a part of the problem, and none of them was found to combine the four functions. The comparison is the project's preliminary assessment and not an independent study of every alternative.

Table 7.1: Existing alternatives and the proposed differentiation

| Alternative | What it addresses | What it does not do for a job seeker |
|---|---|---|
| Manual reporting and word of mouth | Warns others after a scam has happened | Is slow and does not check a new advertisement |

| Alternative | What it addresses | What it does not do for a job seeker |
|---|---|---|
| Browser protection such as Google Safe Browsing | Flags known malicious or phishing websites | Does not judge a job advertisement, an employer's registration or a conversation |
| Sri Lanka Bureau of Foreign Employment portal | Information on licensed foreign-employment agencies | Does not cover local employers or messages, and does not rank jobs |
| Email and message spam filters | Filter bulk unwanted messages | Do not identify recruitment-specific manipulation tactics |
| Review and reputation services (Glassdoor, ScamAdviser) | Employee reviews; website trust scores | Not tied to Sri Lankan registers or to a job ranking |

### 7.2.2 Differentiation

The differentiation is the combination of four functions in one product: multilingual advertisement detection, employer assessment from Sri Lankan registers, tactic-level message analysis and risk-aware ranking. The technical evidence for each function is preliminary. The project has not shown through a market study that it outperforms competing services in practice, and this should be tested with comparative studies and user research.

## 7.3 Business model and feasibility

### 7.3.1 Possible business models

No cost estimates, revenue forecasts, or financial calculations have been made, so feasibility can be discussed only qualitatively. The prototype is a basis for exploring the business models in Table 7.2, none of which has been tested with customers.

Table 7.2: Possible business models (not yet tested)

| Model | Description | Main open question |
|---|---|---|
| Freemium for job seekers | Basic safety features free; advanced services in a paid tier | Willingness to pay |
| Business-to-business | Recruitment platforms or employers pay for the fraud-assessment service or an integration | Which platforms would adopt it |
| Partnerships | Universities and employment-support organizations provide access | Agreements and support |

### 7.3.2 Costs and feasibility

Costs would include hosting the applications and services, databases, model inference, updates of the models as fraud patterns change, dataset preparation, security,

monitoring, technical support and user onboarding. External registers or services may add access costs depending on their terms. Before choosing a model, the team should estimate the cost of a limited pilot, identify the customer groups and test the willingness to pay. Until then, the system is a research prototype with commercial potential and not a financially validated product.

## 7.4 Adoption intellectual property and compliance

### 7.4.1 Ownership and licensing

Commercial adoption would raise questions of ownership, licensing, data protection and responsibility. No decision has been recorded on who owns the intellectual property of the code, models, datasets and documents, or on the license under which the system would be distributed, and this report does not claim exclusive rights for the team, the university or any other party. Before launch, these should be clarified, together with the licenses of third-party libraries, pretrained models, datasets and external sources, and any university requirements.

### 7.4.2 Data protection and responsible use

Job advertisements, employer details and messages can contain personal or confidential information, and Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [16], governs the processing of personal data. A production version should set rules for collection, retention, access and deletion. Component 2 stores no user data and keeps no request history, but the platform as a whole has not been audited against the Act. Risk scores should be presented as estimates based on available evidence and not as legal findings, because an error can harm a genuine business. A registry gap, a classification error or a mistaken report can produce a wrong result, so users should see the reasons, and a way to review or report a result should exist before release.

## 7.5 Sustainable Development Goal contribution

### 7.5.1 Contribution to SDG 8

The project is most directly aligned with SDG 8, and its Target 8.5 on productive employment and decent work [10]. The contribution is indirect: the system does not

create jobs, but it can reduce the loss of money, time and trust that fraud causes during a job search, and it supports Sinhala readers and young job seekers. The experiments and the four-person acceptance study give initial technical and usability evidence and do not show a measurable effect on SDG indicators.

### 7.5.2 Possible negative effects and measures

There are possible negative effects. A false alarm can hurt a genuine business or discourage a job seeker from a real offer; Sri Lankan SMEs that are not in the registers are most exposed, which is why Component 2 describes a missing entry as “not confirmed” and not as fake. Users may rely too much on a risk band, and languages other than English and Sinhala are not served. Measures to sustain the benefit include showing reasons with every risk value, reviewing Medium-band cases, updating the models as scams change, adding Tamil, and measuring in a pilot whether users make better decisions.

# 8 CONCLUSION AND FUTURE DIRECTIONS

## 8.1 Overall conclusion

Online recruitment fraud can occur at different stages of the job-seeking process, from misleading job advertisements and questionable employer information to scam communications that use manipulation tactics. This research addressed the problem by developing an AI-Powered Job Scam Detection Ecosystem that combines four complementary components: multilingual fake job post detection, employer legitimacy verification, scam communication detection, and risk-aware job recommendation. The system brings these functions together through a recruiter web application, a job-seeker mobile application, and supporting backend services.

The evaluation provided evidence of the technical capabilities of the individual components within their respective test datasets. The recommendation evaluation also showed that incorporating fraud-risk information-maintained Precision@10 and MRR@10, slightly improved AP@10 and NDCG@10, and reduced average risk and unsafe rate compared with the baseline. User acceptance feedback from two recruiters and two job seekers indicated that the applications and fraud-awareness features were useful and generally understandable, while also identifying opportunities to improve user guidance and risk explanations. These findings provide preliminary support for the proposed integrated approach. However, the limited user evaluation, dataset-specific results, and absence of established real-world fraud-prevention outcomes mean that the ecosystem's effectiveness in live recruitment environments remains to be validated.

## 8.2 Principal integrated contribution

### 8.2.1 Integration of four components

The principal contribution is the integration itself: four complementary fraud-assessment and ranking functions that share a job record, a common risk scale and an end-to-end workflow, delivered in two applications.

### 8.2.2 Further contributions

Beyond the components, the team produced three things that matter outside this project. First, a working design for passing explainable risk values from several detectors into a multi-criterion ranking (EWM and TOPSIS) so that safety changes what a job seeker sees. Second, evidence about Sri Lankan sources: employer registers give a weak statistical signal on their own, reputation evidence is stronger, and a combination with visible weights costs some recall compared with the best weights. Third, an honest account of where an integrated system of this kind is weak: at the boundary between its parts, in the Medium band, in failure handling and in how users read the explanations.

## 8.3 Future directions

### 8.3.1 Priorities

Table 8.1 lists the future work in order of priority.

Table 8.1: Future directions in priority order

| Priority | Action | Component | Reason |
|---|---|---|---|
| 1 | Correct the three known defects of the employer service; remove the default access key; pin library versions | 2 and all | Needed before any release and for the registration rule to be safe |
| 2 | Handle the Medium band: provisional verification, “not confirmed” note, stored evidence report, later review or sampling | 2, platform | Removes the main safety gap of automatic verification |
| 3 | Define and test failure behavior of the whole system, including a ranking fallback when a detector is down | All | No system-wide fallback exists |
| 4 | Evaluate on larger, real, confirmed data; repeat with several splits; report final results of the post detector on the 4,590-post dataset | 1, 2, 3 | Datasets are limited and partly generated |
| 5 | Larger and more varied user study with clearer risk explanations, after ethical clearance | All | Only four participants so far |
| 6 | Improve FOMO detection; re-evaluate the ranking with more users and a significance test | 3, 4 | Weakest tactic; small ranking gains |
| 7 | Load, security and accessibility testing; monitoring; backups | All | Production readiness |
| 8 | Tamil support, more employer sources or published registry interfaces, a pilot with a platform, and decisions on ownership, licensing and data protection | All | Adoption and commercialization |

### 8.3.2 Closing statement

In conclusion, the project provides an integrated technical foundation for fraud-aware online recruitment. The next stage should strengthen the evidence in real conditions and prepare the product for responsible release.

# REFERENCES


[1] S. Vidros, C. Kolias, G. Kambourakis, and L. Akoglu, “Automatic detection of online recruitment frauds: Characteristics, methods, and a public dataset,” Future Internet, vol. 9, no. 1, art. 6, 2017, doi: 10.3390/fi9010006.

[2] G. Anagha, V. D. Agrawal, G. Sarkar, K. Vemuri, and S. K. Shukla, “Modeling behavioral signals in job scams: A human-centered security study,” arXiv preprint arXiv:2601.19342, 2026.

[3] S. Dutta and S. K. Bandyopadhyay, “Fake job recruitment detection using machine learning approach,” Int. J. Eng. Trends Technol., vol. 68, no. 4, pp. 48–53, 2020.

[4] B. Chiraratanasopha and T. Chay-intr, “Detecting fraud job recruitment using features reflecting from real-world knowledge of fraud,” Curr. Appl. Sci. Technol., vol. 22, no. 6, 2022, doi: 10.55003/cast.2022.06.22.008.

[5] S. Mahbub, E. Pardede, and A. S. M. Kayes, “Online recruitment fraud detection: A study on contextual features in Australian job industries,” IEEE Access, vol. 10, pp. 82776–82787, 2022, doi: 10.1109/ACCESS.2022.3197225.

[6] B. Alghamdi and F. Alharby, “An intelligent model for online recruitment fraud detection,” J. Inf. Secur., vol. 10, no. 3, pp. 155–176, 2019, doi: 10.4236/jis.2019.103009.

[7] T. Pangare, S. Surve, K. Salokhe, and S. Hasbe, “JobScamShield – Fake job post detection,” Int. J. Multidiscip. Res. (IJFMR), vol. 8, no. 2, 2026.

[8] S. Winardi, S. Megawan, N. P. Wong, R. Kurniawan, F. A. Putra, and C. Cynthia, “Utilizing TF-IDF content-based filtering for job recommendation systems,” J. Nas. Komputasi Teknol. Inf., vol. 8, no. 5, pp. 2838–2845, 2025, doi: 10.32672/jnkti.v8i5.9686.

[9] N. S. Honmane, S. Khanorkar, K. Singh, P. Kavhar, and H. Kaware, “Integrated machine learning framework for layoff risk assessment and job recommendation,” in Proc. Int. Conf. Responsible, Risk-Aware, and Regulated AI (RRRAI 2026), Atlantis Press, 2026, pp. 279–290, doi: 10.2991/978-94-6239-723-1_26.

[10] United Nations, “Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.” [Online]. Available: https://sdgs.un.org/goals/goal8 (accessed Oct. 10, 2026).

[11] A. Conneau et al., “Unsupervised cross-lingual representation learning at scale,” in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics, 2020, pp. 8440–8451.

[12] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why should I trust you?”: Explaining the predictions of any classifier,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining, San Francisco, CA, USA, 2016, pp. 1135–1144.

[13] P. Probst, M. N. Wright, and A.-L. Boulesteix, “Hyperparameters and tuning strategies for random forest,” WIREs Data Min. Knowl. Discov., vol. 9, no. 3, art. e1301, 2019.

[14] A. Ghourabi, “SM-Detector: A security model based on BERT to detect SMiShing messages in mobile environments,” Concurrency Comput. Pract. Exper., vol. 33, no. 24, 2021.

[15] A. Shinde, E. Q. Shahra, S. Basurra, F. Saeed, A. A. AlSewari, and W. A. Jabbar, “SMS scam detection application based on optical character recognition for image data using unsupervised and deep semi-supervised learning,” Sensors, vol. 24, no. 18, art. 6084, 2024.

[16] Parliament of the Democratic Socialist Republic of Sri Lanka, Personal Data Protection Act, No. 9 of 2022, Colombo, Sri Lanka, 2022.

# APPENDICES

Appendix A gives the interface of Component 2 and the risk hand-over. Appendix B gives the contribution declaration. Both are cited in the text (Sections 4.2 and 4.5).

## Appendix A Integrated interface specification

Table A.1 gives the interface of Component 2, the employer legitimacy service, which is the interface used both for the public company check and for the registration check. It is a FastAPI service, started with Uvicorn on its own port, with the request and response defined by typed models.

Table A.1: Component 2 interface

| Item | Specification |
|---|---|
| Endpoint | POST /predict (JSON) |
| Request | company_name, website_url, email; all optional; at least a name and a website are advised |
| Response: model part | prediction, probability, confidence (high or low), warning |
| Response: score and band | risk_score (S, 0 to 100), risk_level (Low, Medium or High), verdict, color, recommendation |
| Response: evidence | evidence, score_breakdown (points of each layer), registration_status and its label, summary and trace, registration_sources, registration_method, reg_name, reg_number, opencorporates_url, slaasmb_sbe_name |
| Response: other | features_used (14 features and signals), elapsed_seconds |
| Bands | Low risk $S \geq 70$; Medium risk $45 \leq S < 70$; High risk $S < 45$ |
| Hand-over | Employer risk $r = 1 - S/100$ is stored with the job and used by Component 4 |
| Registration rule | Low and Medium: verified automatically; High: sent to an administrator with the evidence report |

For the other components, the content that crosses each interface is summarized in Table 4.2; their detailed request and response models are in the individual theses.

## Appendix B Team contribution declaration

Table B.1: Contribution declaration

| Member | Individual component | Individual contribution | Shared contribution |
|---|---|---|---|
| Santhosh S (IT22113740) | Multilingual fake job post detection | Dataset, model, explanation and evaluation of Component 1 | Common report |
| Bandara H M N T (IT22251664) | Employer legitimacy verification | Three evidence layers, Random Forest, fusion, evidence report and tests of Component 2 | Common report |
| Kajanthan U (IT22224002) | Scam communication detection; team leader | BSTD model, tactic detection and evaluation of Component 3 | Common report; coordination of the integration across the four components |
| Methmi N P C S (IT22188618) | Risk-aware job recommendation and ranking | TF-IDF relevance, EWM, TOPSIS and evaluation of Component 4 | Common report |

Use of AI tools. The team used a generative AI assistant (Claude, by Anthropic) to help draft and edit parts of this report from the team's own results. The members checked the figures against the project outputs and remain responsible for the content.